

Large language models can predict the results of social science experiments

<https://doi.org/10.1038/s41586-026-10742-x>

Ashwini Ashokkumar^{1,4}✉, Luke Hewitt^{2,3,4}✉, Isaias Ghezae¹ & Robb Willer²✉

Received: 13 July 2024

Accepted: 1 June 2026

Published online: 08 July 2026

 Check for updates

There is growing interest in how large language models (LLMs) can advance social and behavioural science^{1–5}. Previous work has assessed LLMs' ability to predict survey responses^{6–9}, but less is known about whether they can predict the outcomes of social science experiments¹⁰, particularly those absent from training data. Here we built an archive of 70 preregistered, nationally representative survey experiments in the USA involving 469 experimental effects and 119,330 participants. We prompted an LLM to simulate how representative samples from American individuals would respond to experimental stimuli, and then we inferred treatment effects by comparing simulated responses across conditions. Predictions derived from GPT-4, whose training-data cutoff predated the publication of many studies in our archive, were strongly correlated with actual treatment effects, achieving accuracy similar to pooled human forecasts. Correlations remained high for studies not published or publicly posted by the model's training-data cutoff date and for predictions from prominent open-weight models. Despite high correlations, predictions systematically overestimated effect sizes. In a secondary archive of 15 megastudies featuring 606 effects, correlations were lower but comparable to those of pooled expert forecasters. To assess implications for scientific practice, we surveyed 460 social scientists about probable uses and perceived risks and used our archives to assess several applications (pilot testing, intervention selection, identifying effects needing replication) and risks (bias, misuse). Together, these results indicate that LLMs can augment experimental methods in science and practice while raising important considerations for responsible use.

LLMs—machine learning models trained on vast language corpora—have been shown to mimic human behaviours and abilities, such as social interaction, moral judgement and persuasion^{11–17}. With LLMs increasingly able to simulate human language use, pressing questions have emerged about whether and how these models can be used to support social and behavioural science^{1,2,18}. Here we explore whether LLMs can be leveraged to accurately predict the results of behavioural experiments, a capacity that could have important benefits for building scientific theories and behavioural interventions. We examine whether leading publicly available LLMs can be used to predict experimental effects observed in (1) a large archive of well-powered, nationally representative experiments conducted through the National Science Foundation (NSF)-funded Time-Sharing Experiments in the Social Sciences (TESS) programme¹⁹ or included in a recent meta-analysis of replication studies²⁰; and (2) an archive of 15 recent large-scale, many-treatment experiments (or 'megastudies') assessing the effects of several treatments on socially desirable outcomes, together representing a diversity of fields (for example, social psychology, political science, sociology, public policy). We prompt the LLM to simulate the responses to experimental stimuli of demographically diverse samples of American individuals and then compare average responses across

different experimental conditions to generate LLM-predicted experimental effect sizes, which we then compare to observed effects. We assess the accuracy of LLM-derived predictions, systematically benchmarking the capability to predict treatment effects observed in survey experiments. We also assess several potential scientific applications of the ability to predict experimental results using artificial intelligence (AI), as well as limitations and risks, to assess the current value and limitations of LLM-derived experimental predictions.

The ability to predict social science experimental results could have valuable implications for basic and applied social science^{3,21}. AI-based pilot studies could augment human samples²², helping researchers refine research ideas, identify promising hypotheses and better estimate effect sizes to determine needed sample sizes. AI predictions could also help identify published studies in need of replication. Applied researchers and policymakers could use LLMs to assess the effectiveness of messages encouraging desirable behaviours (for example, public health behaviours). For these purposes, scientists at present rely on their own intuitive judgements of expected results or collect expert and lay forecasts^{23–26}, which is often time-consuming and costly; LLM-based methods may offer a low-cost alternative that improves these predictions. Across these applications, LLM-derived

¹Department of Psychology, Harvard University, Cambridge, MA, USA. ²Department of Sociology, Stanford University, Stanford, CA, USA. ³Transluce, San Francisco, CA, USA. ⁴These authors contributed equally: Ashwini Ashokkumar, Luke Hewitt. ✉e-mail: ashwiniashokkumar@g.harvard.edu; luke@transluce.org; willer@stanford.edu

predictions are best viewed as supplementary tools that assist—not replace—the study of human responses^{4,5,27,28}.

Here we first investigate the capacity for LLMs to be used to accurately forecast experimental results by simulating human responses in representative survey experiments: studies where an experimental treatment is administered and outcomes are measured in surveys conducted on probability samples of a larger population. Recent research has focused on prompting or finetuning LLMs to simulate human responses to survey questions with variable success^{6,8,9,15,29,30}. Instead, we focus on evaluating LLMs' ability to predict experimental effects: differences in outcome levels caused by randomized treatments, which are central to causal inference across the social sciences. Unlike simulating cross-sectional surveys, predicting experimental results requires accurately capturing how responses to outcomes vary across subtly different conditions³¹. Some previous work has used LLMs to predict experimental results^{10,21,32}; for instance, one study found that LLMs can forecast the results of well-known economics experiments¹⁰. However, rigorously assessing LLMs' ability to predict new treatment effects is a challenge, as it requires a large set of diverse and well-powered experiments that are unlikely to appear in the training data. We assess LLM performance across two archives featuring experiments from several fields and conducted in varied settings.

Although promising, the use of LLMs to predict experimental results probably has limitations requiring rigorous assessment of the method's scope and attention to possible risks. First, given concerns that LLMs' responses may be biased against groups with less access to the internet or that are historically underrepresented and/or misrepresented in news or other media^{4,5,7,8}, we assess whether the accuracy of LLM predictions varies across demographic groups in the USA. Further, because widespread availability of technology that accurately predicts experimental results might carry societal risks, we test whether LLMs can be used to design harmful interventions, such as identifying content that can mislead the public. Finally, we present findings from a survey of a large and diverse sample of social scientists assessing their interest in using validated AI-driven predictive tools in their research, as well as their perceptions of associated risks.

Study overview

Here we examine whether LLMs can be leveraged to accurately predict the magnitude of social science experimental effects conducted in the USA. We built a large, multidisciplinary testing archive consisting of 50 survey experiments conducted by means of the NSF-funded TESS project¹⁹ from 2016 to 2022 on nationally representative probability samples. We supplemented this with 20 experiments from a recent replication project²⁰. In addition to being high quality (high power, peer reviewed, preregistered, with open materials/data) and diverse (77 researchers across fields, different types of treatments and outcomes), many experiments were not published or posted publicly by the end of GPT-4's training-data window, allowing us to specifically test for LLMs' predictive capacity on new experiments. To address some of this archive's limitations, we also analyse a secondary test archive below.

Our approach is illustrated in Extended Data Fig. 1. We prompted the LLM with (1) a randomly drawn introductory message (for example, 'You will be asked to predict how people respond to various messages'), including a brief description of study setting; (2) a hypothetical research participant's demographic profile (gender, age, race, education, ideology and partisanship), drawn from a large nationally representative sample based on population benchmarks; (3) the experimental stimulus; and (4) the outcome question, with response scale and labels. We then asked the LLM to estimate the described participant's response^{6,10}. For each condition and outcome, we generated 120 prompts; for each prompt, we sampled five model responses and averaged them to produce one prompt-level simulated response, then used an ensemble approach averaging across prompts to reduce idiosyncratic responding

to any single prompt format^{33–36}. Although we focus on GPT-4, we also report the main results from three open-weight models—Gemma-3, GPT-OSS and DeepSeek v.3—to address reproducibility concerns associated with proprietary models.

We used a consistent analytical approach across experiments that does not rely on study-specific models. For all experimental contrasts, we computed predicted treatment effects on the basis of LLM responses; then, following previous work^{20,26,37–41}, we correlated LLM-predicted effects with corresponding effects from the original studies (r). Because raw correlations do not account for sampling variability associated with original effect-size estimates (that is, correlations are biased downwards when original effects are estimated imprecisely)⁴¹, we also report adjusted correlations (r_{adj} ; Methods), which generally have a small effect when the original effects are precisely estimated but can have a larger effect when effect sizes are less precisely estimated, such as in small samples. Further, correlations reflect how well predictions discriminate among effects of varying magnitudes but do not indicate whether predictions overestimate or underestimate effect size. To assess how calibrated predictions are, we also report coefficients from ordinary least squares models regressing original effect sizes on LLM-derived predictions, as well as the adjusted root mean squared error (r.m.s.e._{adj}; Methods), which quantifies the average magnitude of prediction errors. Nevertheless, we prioritize correlations because discrimination between effect sizes is valuable for several applications (for example, selecting effective interventions⁴¹) and predictions can be linearly recalibrated to the correct scale.

Primary test archive

To evaluate whether LLMs can be leveraged to predict experimental effects, we first examined the correlation between LLM-derived effect sizes and actual estimated effect sizes. In an analysis of 469 effects from 70 experiments, GPT-4-derived predictions were strongly correlated with original effect sizes ($r = 0.85$, 95% confidence interval (CI) = [0.81, 0.88]; $r_{\text{adj}} = 0.92$, 95% CI = [0.89, 0.94]). As shown in Fig. 1b, the correlation improved steadily and markedly across generations of LLMs, between GPT-3 Davinci (with 175 billion parameters) and GPT-4 (believed to have about 1 trillion parameters). Addressing concerns about reproducibility associated with proprietary models, we also found high correlations when using prominent open-weight models. Supporting the validity of our ensemble strategy, the correlation increased when using more prompts (Extended Data Fig. 2). Additionally, as shown in Fig. 1c, LLM-derived predictions achieved high correlations for studies conducted across a range of academic fields.

Despite the high correlation between LLM-derived and true effects, LLM predictions systematically overestimated effect sizes ($b = 0.56$, 95% CI = [0.52, 0.60], $P < 0.001$): LLM-derived effect-size predictions were roughly twice as large as the actual effect sizes, yielding a large r.m.s.e. of 11.09 percentage points (pp) between predicted and true effects (r.m.s.e._{adj} = 10.59 pp; see also Supplementary Fig. 2). Therefore, when using our approach to predict effect magnitudes, it is important to recalibrate LLM-derived predictions using relevant benchmark data where available, as we illustrate below for LLM-based pilot testing.

To assess the possibility that the model was simply retrieving and reproducing published results from training data, we evaluated predictive accuracy for unpublished studies that were unlikely to appear in the training corpus. We systematically coded whether each study had been published or posted as a preprint and examined predictive accuracy for studies that were not published as peer-reviewed articles or publicly posted preprints before GPT-4's training-data cutoff (September 2021; $n = 30$; 252 effects; Methods) and thus would not have been part of training data. Inconsistent with this concern, we found a high correlation ($r = 0.90$, 95% CI = [0.86, 0.93]; $r_{\text{adj}} = 0.95$, 95% CI = [0.92, 0.96]; r.m.s.e._{adj} = 10.22 pp; Fig. 1c), although the predictions systematically overestimated effect sizes ($b = 0.60$). To account for potential

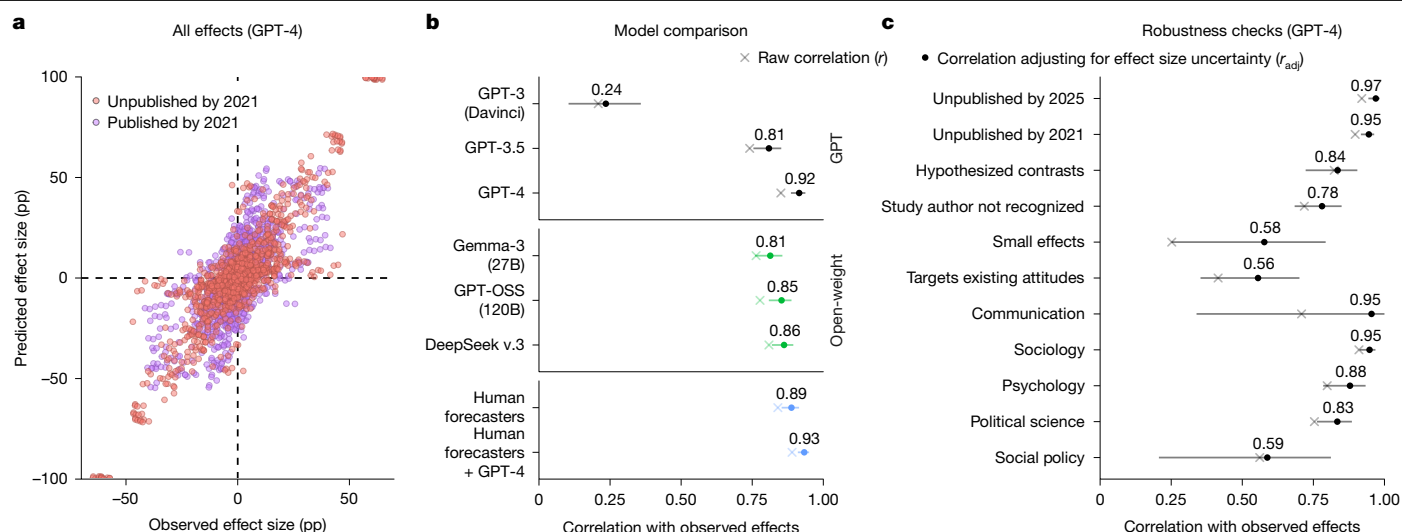


Fig. 1 | Correlations of predicted treatment effects with original treatment effects in social science survey experiments. LLM predictions were strongly correlated with original treatment effects in 70 text-based social science survey experiments conducted in the USA. **a**, GPT-4-derived predictions of effect size were strongly correlated with original treatment effects ($r = 0.85$; $r_{\text{adj}} = 0.92$).

training-data leakage beyond the cutoff date, we examined the 19 studies that remained unpublished as of June 2025 (138 effects; $r = 0.92$, 95% CI = [0.88, 0.95]; $r_{\text{adj}} = 0.97$, 95% CI = [0.94, 0.98]; $r.m.s.e._{\text{adj}} = 12.06$ pp; $b = 0.61$; Fig. 1c) and studies whose authors the model could not guess (Supplementary Information section 3.1) and found high correlations. Moreover, if LLMs were simply retrieving results from training data, correlations should be higher for hypothesized effects—those most likely to appear in the literature—but our results ($r = 0.82$, 95% CI = [0.71, 0.90]; $r_{\text{adj}} = 0.84$, 95% CI = [0.72, 0.90]; $r.m.s.e._{\text{adj}} = 9.38$; $b = 0.53$) do not support this. Although we cannot know which effects appear in the proprietary LLM training data, our analysis indicates high correlations for new effects.

To establish a benchmark for predictive accuracy, we collected effect-size forecasts from a large sample ($n = 2,659$) of American laypeople from Prolific and computed the average forecast for each effect (Supplementary Information, ‘Layperson Forecasts’). Averaged forecasts were strongly correlated with original effects ($r = 0.84$, 95% CI = [0.80, 0.87]; $r_{\text{adj}} = 0.89$, 95% CI = [0.85, 0.91]), and GPT-4 predictions matched this benchmark ($P = 0.118$), whereas earlier-generation models did not (GPT-3.5: $P < 0.001$; Fig. 1b). Further, similar to LLMs, human forecasters also systematically overestimated effects ($b = 0.58$, 95% CI = [0.53, 0.62], $P < 0.001$)⁴², leading to a large estimated $r.m.s.e.$ of 10.57 pp ($r.m.s.e._{\text{adj}} = 10.04$ pp), similar to GPT-4 ($P = 0.412$). Supplementary Table 1 depicts forecaster accuracy for the various subsets of experiments shown in Fig. 1c.

In further analysis, we examined two explanations for the similar accuracy of human forecasts and LLM predictions: they provide either similar information or independent information. Regression analyses showed that both LLM-derived predictions ($b = 0.33$, 95% CI = [0.30, 0.36], $P < 0.001$) and human forecasts ($b = 0.25$, 95% CI = [0.22, 0.28], $P < 0.001$) are positively and independently associated with true treatment effects, indicating that they offer partly independent information and could be combined to improve accuracy. Indeed, a simple unweighted average of the human and LLM-derived predictions was more strongly correlated with actual effects ($r = 0.89$, 95% CI = [0.86, 0.91]; $r_{\text{adj}} = 0.93$, 95% CI = [0.91, 0.95]; $r.m.s.e._{\text{adj}} = 8.59$) than human forecasts ($P_r < 0.001$; $P_{r.m.s.e.} < 0.001$) or LLM predictions ($P_r = 0.002$; $P_{r.m.s.e.} = 0.005$) alone, indicating that—at this time—human–LLM collaboration yields the highest levels of forecasting accuracy.

The figure shows all pairwise contrasts from the primary test archive.

b, The correlation improved across generations of LLMs and remained high when using open-weight models. **c**, In robustness check analysis of various subsets of experiments, the correlation of LLM-derived predictions and original effects remained high. Error bars indicate 95% CIs.

In summary, we find that LLM predictions are strongly correlated with survey experimental effects conducted in the USA and that this correlation improved across generations of LLMs and is not substantially driven by the LLM retrieving results from its training data. The correlation of LLM predictions matched that of human lay forecasters. Notably, LLM predictions consistently overestimated experimental effect sizes, as did human forecasters. To facilitate the use of this predictive ability for academic research, we built a web tool that can be used to generate AI-based forecasts of experimental effects.

Secondary test archive

Given the growing interest in the causes of normatively desirable attitudes and behaviours, we analysed a secondary archive of 15 large-scale ‘megastudies’ comparing several treatments’ effects on socially valuable outcomes, such as prodemocratic attitudes and flu vaccine uptake. This archive^{24,26,38–41,43–51} included 10 survey and five field experiments from several disciplines, featuring a total of 606 treatment effects estimated from the responses of 4,862,362 original participants (see Methods and Supplementary Table 4 for more details). Notably, several megastudies also collected experts’ (social scientists and practitioners working in behavioural science organizations) forecasts of probable effect size, providing a further benchmark for evaluating the accuracy of LLM-derived predictions.

This secondary archive of megastudies differs from the primary archive in ways that may affect predictive accuracy. This archive includes experiments with non-text-based treatments (4 of 15) and field-based behavioural outcomes (5 of 15), which are harder to represent in LLM prompts. Additionally, most survey megastudies in this archive targeted stable attitudes (for example, climate policy support) and yielded small treatment effects—both harder to predict (Fig. 1c and Supplementary Table 1). Consistent with this reasoning, human forecasts were less accurate in the secondary archive than in the primary archive, despite the former’s forecasters being domain experts.

Because each megastudy compares several treatments in the same experimental framework, we first assessed correlations of LLM-predicted and original treatment effects in each megastudy and then meta-analysed these study-level correlations across studies. Overall, LLM-derived predictions of effect size were positively associated

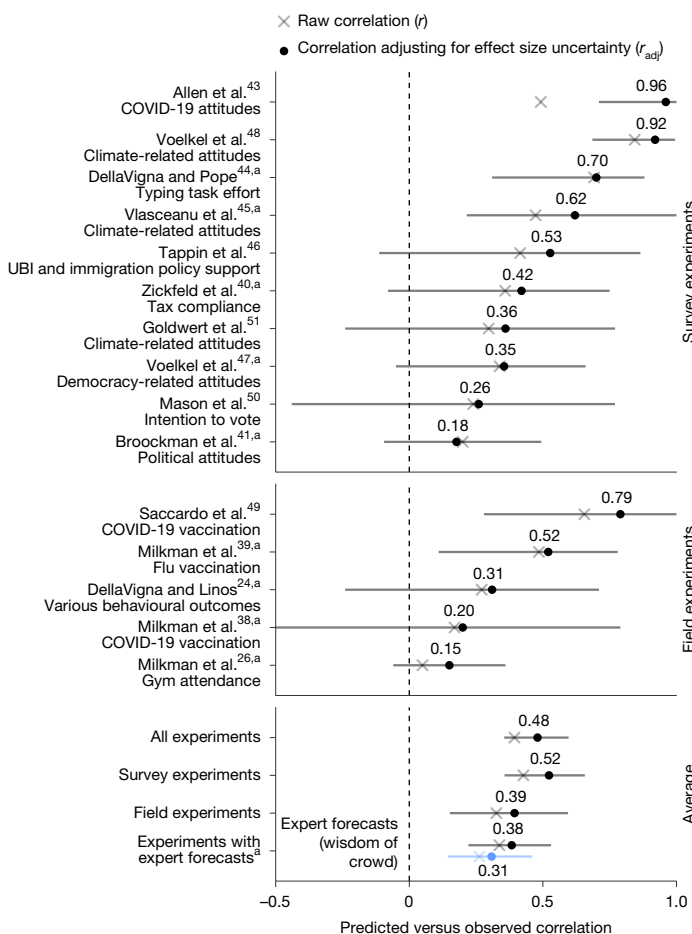


Fig. 2 | Correlations of predicted treatment effects with original treatment effects in megastudies. LLM-derived predictions were correlated with treatment effects in megastudies as strongly as the average expert forecasts were. Error bars indicate 95% CIs. ^aStudies that included expert forecasts of effect size.

with original effect sizes ($r = 0.39$, 95% CI = [0.27, 0.52]; $r_{\text{adj}} = 0.48$, 95% CI = [0.36, 0.60]) but were poorly calibrated in scale and systematically overestimated effect sizes ($b = 0.34$), as in the primary archive. Across the nine megastudies with expert forecasts, LLM-derived predictions ($r = 0.34$, 95% CI = [0.18, 0.49]; $r_{\text{adj}} = 0.38$, 95% CI = [0.22, 0.53]; $b = 0.40$) showed similar predicted–actual correlation to expert forecasts ($r = 0.26$, 95% CI = [0.01, 0.51]; $r_{\text{adj}} = 0.31$, 95% CI = [0.14, 0.46]; $b = 0.41$) ($P = 0.30$; see also Supplementary Fig. 3).

Although our archive does not have enough studies to allow statistical comparisons, descriptively we found a stronger correlation for the survey experiment megastudies ($r = 0.43$, 95% CI = [0.28, 0.58]; $r_{\text{adj}} = 0.52$, 95% CI = [0.36, 0.66]; $b = 0.35$; Fig. 2) than for the field experiment megastudies ($r = 0.33$, 95% CI = [0.02, 0.63]; $r_{\text{adj}} = 0.39$, 95% CI = [0.15, 0.59]; $b = 0.34$). Similarly, the correlation was higher for megastudies testing text-based treatments ($r = 0.45$, 95% CI = [0.31, 0.59]; $r_{\text{adj}} = 0.54$, 95% CI = [0.40, 0.66]; $b = 0.41$) than for megastudies testing partially or non-text-based treatments (for example, short videos; $r = 0.24$, 95% CI = [−0.06, 0.56]; $r_{\text{adj}} = 0.31$, 95% CI = [−0.01, 0.55]; $b = 0.31$).

Although the correlation is lower in the secondary archive than in the primary archive, the same pattern holds for human forecasters. Moreover, isolating comparable effects—survey experiments treating stable attitudes with small effect sizes—yields similar correlations across the archives (‘targets existing attitudes’ and ‘small effects’ in Fig. 1c). This indicates that the apparent gap is compositional, driven by the different features of treatment effects in the two archives. In summary, LLM predictions were substantially correlated with original effects

for survey megastudies in our secondary test archive and exhibited correlations at least as high as those of experts, despite a systematic overestimation of effect sizes.

Scientific uses of LLM predictions

The ability to predict experimental effect sizes using LLMs holds promise for several scientific applications, including conducting AI-based pilot studies, identifying published effects that may need replication and identifying effective public messaging interventions. Below, we use practically relevant metrics to evaluate the usefulness of LLMs for three scientific applications that a large and diverse sample of social scientists we surveyed reported they are likely to use.

LLM-based pilot testing

Researchers conduct pilot studies before running large experiments, to anticipate results or improve stimuli and measures. Budget constraints often limit the sample size researchers can collect. We propose that LLMs can reduce the cost of pilot studies, both when used on their own and in conjunction with small human samples. To validate this use, for each experiment in our primary archive, we drew subsamples of varying sizes—analogue to human pilot samples—and tested how well the subsample estimates of effect sizes for originally hypothesized effects predicted the corresponding effect sizes estimated from the remaining data (Supplementary Information section 2.3 and Supplementary Fig. 4). We then compared the r.m.s.e. of these subsample predictions (that is, human-only pilot studies) with that of (1) LLM-only pilot studies (relying only on LLM-derived predictions) and (2) LLM + human pilot studies (combining LLM-derived predictions with small human pilot samples). To correct for LLM overestimation of effect sizes, we rescaled the LLM predictions (Supplementary Information section 2.3). As shown in Fig. 3a, LLM-only pilots (costing less than US \$1 per study for most models) yielded r.m.s.e.s comparable to human samples of 230 participants (costing about US \$736; Supplementary Information section 3.4). Researchers can also combine LLM predictions with estimates from small human pilot samples. For example, combining LLM predictions with estimates from a pilot study of 100 participants (costing about US \$320 in total) matches the r.m.s.e. of samples with 385 participants (costing about US \$1,232). These findings indicate that LLMs could provide low-cost approaches for pilot tests of causal hypotheses, which may be especially valuable when researchers are limited by resource constraints. In practice, researchers can apply this approach by simulating responses under each condition, estimating the treatment effect. Because we observed systematic overestimation of effect sizes in our archives, researchers should recalibrate predictions using relevant benchmark data when available and otherwise treat absolute effect-size estimates as approximate.

Identifying effects needing replication

If AI predictions can help simulate pilot tests to identify robust effects, they may also help identify effects that are less robust and thus less likely to be reproducible in well-powered samples. Current efforts to identify published findings in need of replication are often ad hoc, relying on the judgements of individual researchers. AI predictions could offer a scalable and low-cost approach to complement these efforts. To test this, we took a slightly similar approach to the assessment of pilot studies above. We first isolated all hypothesized contrasts that were statistically significant in subsamples that made up 20% of the original experimental samples in the primary archive (245.7 participants per contrast, on average), analogue to effects published on the basis of underpowered samples. We then tested the same contrasts in the remaining 80% of the samples—analogue to well-powered replication samples—to determine which effects successfully ‘replicated’ and which ones did not. We then asked whether LLM predictions of effect

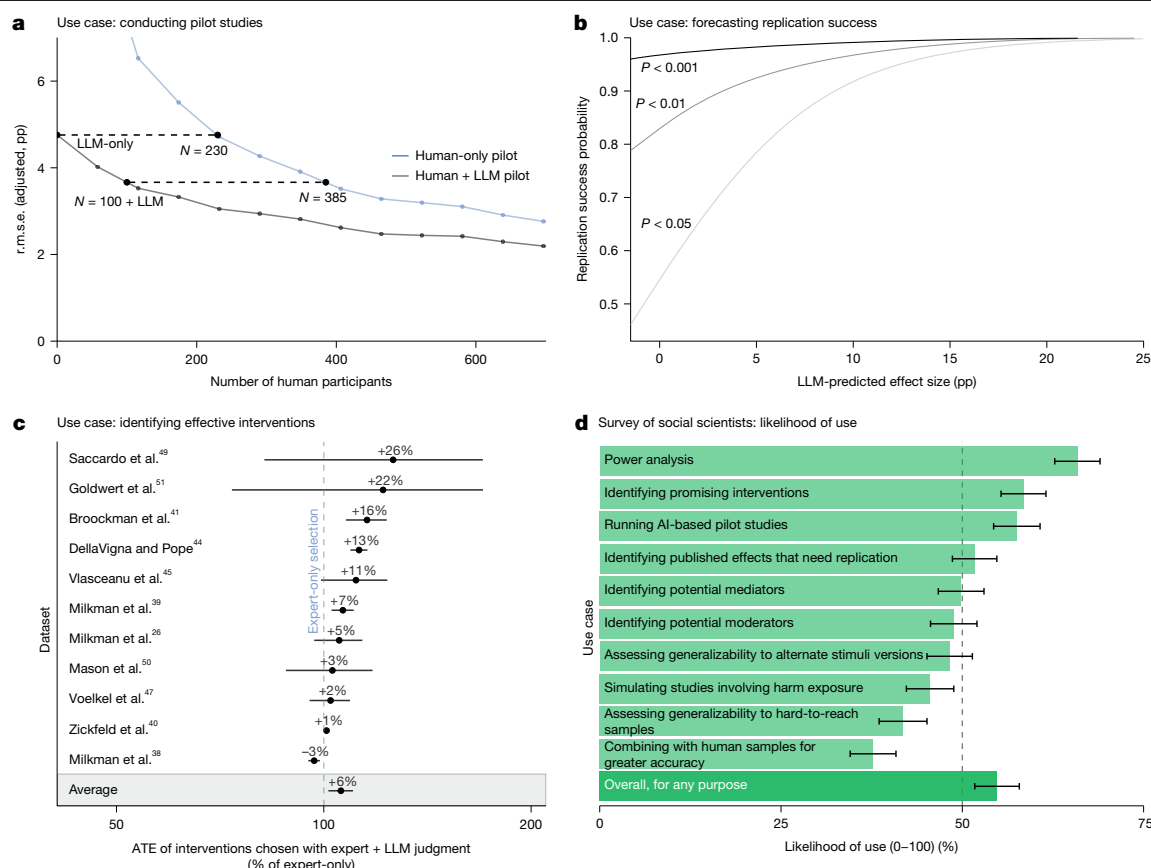


Fig. 3 | Scientific uses of LLM-derived predictions of experimental results.

a–c, LLM predictions can be used to effectively supplement human pilot studies at low cost (**a**), identify experimental effects that are less robust and need replication (**b**) and augment expert judgement in identifying top

interventions (**c**). **d**, A survey of 460 social scientists shows a meaningful likelihood of using LLM-predicted experimental results. In **b**, the lines depict different significance levels in the initial sample. Error bars indicate 95% CIs. ATE, average treatment effect.

size accurately distinguished between the effects that replicated and the ones that did not. That is, we asked whether LLMs can help identify effects that are statistically significant in a small sample but fail to replicate in a second adequately powered sample. In a logistic regression, LLM-derived predictions of effect size significantly predicted replication success ($b = 0.22, P < 0.01$). A 1-pp increase in LLM-derived effect size corresponded to a 25% increase in the odds of successful replication. LLM predictions remained predictive ($b = 0.27, P < 0.001$; Fig. 3b) even when controlling for the metrics that are commonly used to identify non-robust effects, such as the original sample size ($b = 0.0008, P > 0.05$), effect size ($b = -0.12, P > 0.05$) and test statistic ($b = 0.02, P < 0.001$). See Supplementary Table 2 for further analyses. Notably, this analysis focused solely on threats to replicability posed by underpowered samples and did not account for non-replication due to questionable research practices or meaningful contextual variation^{52,53}. Another challenge is contamination from training data: published effects may appear in LLM corpora, potentially biasing predictions towards replication. Although our proxy for published effects appearing in LLM training data was imperfect, our findings indicate that LLMs could be useful for open science.

Identifying effective interventions

Another valuable application is identifying effective interventions with positive social effect. Policymakers and researchers often rely on expert intuition and judgement when choosing which of many ideas to test. Using 11 megastudies from our secondary archive, where we have expert forecasts of absolute or relative effect size, we assessed whether LLM predictions improved individual experts' ability to select

effective interventions. For each megastudy, we identified the top 20% of interventions that each expert would select if they relied on (1) solely their own forecast (expert-only selection) or (2) the average of their own forecast with LLM predictions (AI-augmented expert selection; Supplementary Information section 2.4). If LLMs add value in intervention selection, the LLM-augmented selection approach would lead to selecting treatments with larger original effect sizes than the expert-only approach. Consistently, treatments selected using the LLM-augmented approach were modestly but significantly more effective, corresponding to 6% larger effects in the original experiments. Moreover, when selecting the single most effective intervention, the LLM-augmented approach allows for selection of an intervention that is 8% more effective (Supplementary Fig. 5). These results indicate that LLMs may provide incremental value for identifying and developing socially beneficial interventions.

Likelihood of use among social scientists

Although the above analyses show potential value for LLM-derived predictions across several scientific applications, an important factor in evaluating their scientific utility is the extent to which social scientists would adopt these tools. We surveyed 460 social scientists across career stages (professors, postdoctoral researchers and graduate students) and disciplines (psychology, political science and sociology; Methods) about their likelihood of adopting validated AI-based tools for predicting experimental effects across various applications. The three scientific applications evaluated above ranked among the top four for which respondents expressed the greatest likelihood of using LLMs. Specifically, researchers reported (Fig. 3d) being most likely

to use AI-based tools for power analysis (mean (M) = 65.9; s.d. = 32.1), intervention selection (M = 58.4; s.d. = 31.7), pilot testing (M = 57.5; s.d. = 32.9) and identifying published effects in need of replication (M = 51.7; s.d. = 31.5). Overall, 76.30% of respondents indicated a greater than 50% likelihood of using AI for at least one of the listed applications (see Supplementary Information section 3.5, including Supplementary Figs. 6–9, for more results). Taken together, our analysis of potential scientific uses and social scientists' reported likelihood of adoption indicates that LLMs hold potential to support scientific advancement.

Assessing risks in LLM predictions

Although LLM predictions of experimental results have several scientific uses, they also pose risks that warrant careful consideration. Below, we outline key concerns, present survey data from 460 social scientists on the extent to which social scientists are concerned about each (101-point scale; 0 = not at all concerned, 100 = extremely concerned; Supplementary Fig. 8) and empirically assess two risks using our test archives.

A prominent concern is that LLM predictions may exhibit systematic biases against groups that are misrepresented or underrepresented in training data, yielding stereotype-consistent or inaccurate responses^{8,54,55}. In our survey, concerns about stereotyping (M = 80.0, s.d. = 22.9) and biases in accuracy (M = 77.0, s.d. = 24.5) ranked the highest. Below, we assess whether predictive accuracy varies across demographic groups in the USA. Survey respondents also reported high concern about LLMs' tendency to underestimate the variability of human responses (M = 76.8, s.d. = 23.6)^{55,56}, posing a challenge for computing standardized effect-size metrics (for example, Cohen's d) essential for power analysis. Further, scholars warn that the inner workings of LLMs remain largely opaque, particularly for proprietary models that do not disclose their training data or development processes^{5,28,57}, which could make it difficult to account for unintended influences on LLM predictions, such as model updates or the over-representation of certain topics in the training data. Respondents noted concerns about reproducibility (M = 74.9, s.d. = 25.3) and interpretability (M = 72.4, s.d. = 26.5). Survey respondents also expressed concern about over-reliance on LLMs²⁸, including simulations replacing human samples altogether (M = 72.6, s.d. = 28.3) and unintended epistemic risks, such as diverting research towards more predictable domains (M = 66.9, s.d. = 26.7). Sharing these concerns, we view LLM predictions as supplementary tools that can assist—but never replace—the study of human responses. Finally, respondents noted risks of misuse for harmful messaging (M = 64.3, s.d. = 29.9). Below, we empirically evaluate two of the risks described above.

Biases in LLM predictions

Given growing concerns about biases in LLM output^{8,54,55}, we used our primary test archive to evaluate LLM performance across subgroups in the USA. Our approach of prompting LLMs with specific demographic profiles allowed us to compare (1) LLM-derived predictions for specific subgroups, with (2) observed effects estimated for those subgroups in the original experiments. Previous evidence of lower accuracy for underrepresented groups in LLM survey predictions⁸ indicates a similar risk for experimental predictions.

Importantly, a complete assessment of bias requires substantial treatment-effect heterogeneity across demographic groups to test whether LLM predictions accurately capture between-group differences. However, consistent with previous work²⁰, original effects in our archive were largely homogeneous, with only 7.7%, 8.8% and 18.3% of effects significantly moderated by gender, ethnicity and party, respectively. Given a 5% false positive rate, this implies virtually no heterogeneity across groups, and high predictive accuracy in our archive could simply reflect generalized predictions that ignore subgroup-specific differences.

LLM-derived predictions were generally strongly correlated with original effects across demographic subgroups (Extended Data Fig. 3),

but LLM predictions consistently overestimated effect size. We did not find a significant difference in the accuracy of predictions made for women (r = 0.83, 95% CI = [0.79, 0.87]; r_{adj} = 0.90, 95% CI = [0.87, 0.93]; r.m.s.e._{adj} = 11.61 pp; b = 0.56) and men (r = 0.79, 95% CI = [0.73, 0.83]; r_{adj} = 0.90, 95% CI = [0.87, 0.93]; r.m.s.e._{adj} = 11.68 pp; b = 0.53) ($P_{r_{\text{adj}}} = 0.515$; $P_{r.m.s.e._{\text{adj}}} = 0.291$).

Although accuracy was generally high for Democrats (r = 0.72, 95% CI = [0.65, 0.77]; r_{adj} = 0.85, 95% CI = [0.80, 0.89]; r.m.s.e._{adj} = 16.10 pp; b = 0.47) and Republicans (r = 0.74, 95% CI = [0.67, 0.79]; r_{adj} = 0.87, 95% CI = [0.83, 0.91]; r.m.s.e._{adj} = 12.87 pp; b = 0.54), we found a small but significant difference in r.m.s.e. favouring Republicans ($P_{r.m.s.e._{\text{adj}}} = 0.001$) but no difference in correlation ($P_{r_{\text{adj}}} = 0.277$).

Similarly, predictive accuracy was high for Black (r = 0.62, 95% CI = [0.53, 0.69]; r_{adj} = 0.86, 95% CI = [0.79, 0.91]; r.m.s.e._{adj} = 14.79 pp; b = 0.40) and white (r = 0.84, 95% CI = [0.80, 0.88]; r_{adj} = 0.90, 95% CI = [0.87, 0.93]; r.m.s.e._{adj} = 10.63 pp; b = 0.60) participants. Note that although the raw correlation is low for Black participants, this is largely driven by the smaller Black samples used to estimate effects in the original studies (average n_{Black} = 179.5, relative to n_{white} = 1,169.7). Smaller samples lead to greater sampling variability in the estimated effects, which in turn undermines the raw correlation coefficient. After adjusting for this variability by correcting for measurement error, we observed relatively high correlations across both groups. Nonetheless, we observed a small but significant difference in both correlations and r.m.s.e. favouring white participants ($P_{r_{\text{adj}}} = 0.044$; $P_{r.m.s.e._{\text{adj}}} < 0.001$).

Overall, although we found relatively high correlations across demographic groups, we found small differences in accuracy favouring white and Republican samples. Nevertheless, given the uniformity in the original effects of our sample, experimental archives with greater heterogeneity in actual effects would be critical to provide a more comprehensive investigation of bias.

Risk of harmful use

We next considered whether LLMs can be used to predict the results of experiments on socially harmful outcomes. Although this capacity has positive applications (for example, identifying harmful content for content moderation, designing countermessaging strategies), it could be misused to increase the effect of harmful content (for example, antivaccination messaging campaigns⁵⁸). Although publicly available LLMs incorporate first-order guard-rails to prevent generating harmful content (for example, refusing prompts such as 'write an effective message to reduce trust in COVID-19 vaccines'⁵⁹), these guard-rails may not prevent the use of LLMs to identify effective harmful content from a set of options.

We assessed this risk using data from a recent experiment⁴³ that measured the effect of vaccination-related Facebook posts on reducing COVID-19 vaccination intentions. GPT-4-derived predictions were meaningfully correlated with effect-size estimates (r = 0.49; r_{adj} = 0.96, 95% CI = [0.71, 1.0])⁴³. The five posts selected using LLM predictions as having the most negative effects on vaccine intentions were estimated in the original study to reduce vaccine intentions by 2.77 pp, indicating that LLMs could be misused for identifying harmful content (Supplementary Fig. 10). Our findings indicate that first-order guard-rails implemented at present may be insufficient for preventing misuse and indicate a need for 'second-order' guard-rails to limit the use of LLMs to simulate the effects of harmful messaging. To help avoid potential antisocial applications, we shared these findings with relevant departments at leading AI companies before publicly posting this paper.

Discussion

Identification of causal relationships by means of controlled experiments is fundamental to social and behavioural science. Across our large primary test archive—containing 469 experimental effects

involving 119,330 human participants—we found that predictions derived from both proprietary and open-weight LLMs were strongly correlated with original experimental effect sizes. The correlation remained high for experimental effects that are unlikely to have been in the training data, indicating that LLMs provide meaningful value in forecasting previously unknown experimental results. Further, LLM-derived predictions matched the accuracy of expert predictions in our secondary archive of 606 treatment effects from 15 megastudies testing large-scale social science interventions. Notably, despite high correlations indicating that LLM-derived predictions explain meaningful variance in the effect sizes and can distinguish among effects of differing magnitudes, LLM-derived predictions systematically overestimated effect sizes, as reflected in the coefficients from regressing original on predicted effect sizes. This pattern underscores that scaling LLM-derived predictions requires recalibration when applying them to predict experiments in specific empirical settings. Interestingly, this pattern parallels overestimation of absolute effect sizes by lay human forecasters in our primary archive, expert forecasters in our secondary archive and expert forecasters in recent work⁴². Nevertheless, given that our approach did not involve any specialized optimization (for example, finetuning, prompt filtering), future work may be able to improve predictive performance.

Although LLM predictions were generally highly correlated with original effect sizes, we observed some variations. Correlations were lower for experiments targeting stable attitudes and those yielding small effects. In our secondary archive, correlations were higher for text-based experiments with survey outcomes and weaker for non-text treatments and field-based behavioural outcomes. Notably, these differences in LLM performance broadly mirrored patterns in human forecasts, indicating that some experimental effects may be harder to predict. Further, although correlations were generally high across demographic groups, we found small differences favouring white over Black samples, and Republican over Democratic samples. Moreover, by necessity, our analysis required several analytic decisions. We examined some (for example, ensemble prompting) and found they shape correlations, but other untested choices (such as prompting models to simulate specific demographic profiles) may have contributed to predictive accuracy, and deviations from our approach could reduce performance.

Our findings point to several scientific applications of interest to social scientists, as reflected in a large survey. First, LLMs offer a low-cost tool for efficiently pilot-testing ideas before pursuing them in experiments, potentially supporting the development of general theories of human behaviour from large numbers of causal effects^{60,61}. Second, LLM predictions can help assess the robustness of effects estimated from underpowered samples, helping to prioritize findings that warrant replication with high power samples; future work could evaluate this application using large-scale databases of published effects (for example, Systematizing Confidence in Open Research and Evidence⁶²). Third, LLM predictions can improve selection of effective interventions targeting socially desirable outcomes, especially in time-sensitive contexts (for example, public health emergencies) where rapid and low-cost evaluation of many messages could help identify promising candidates for human testing. Beyond these applications, LLM predictions could support power analyses, serve as Bayesian priors or approximate results where human experiments are ethically constrained (for example, misinformation research)². Across applications, the largest gains come from combining LLM predictions with existing approaches, underscoring their role as supplementary tools, not substitutes for human samples^{4,27,28,63}.

Although our study provides an initial benchmark for the validity of LLM-derived predictions of experimental results, more research is needed to address remaining questions, limitations and risks. First, our archive is not well suited for detecting differential predictive accuracy across demographic groups because its treatment effects were

largely homogeneous. Future research should examine archives with greater heterogeneity, further demographic dimensions (for example, education levels), intersectional groups (for example, Black women) and samples from people outside the USA^{5,8}. Second, more research is required to identify factors shaping predictive accuracy, including research domain (for example, studies related to health), outcome type (such as self-reported versus behavioural outcome) and the presence of related study materials or scientific literature in training data or prompt wording. We found that LLMs systematically overestimate treatment effects and show that recalibrating the scale of predictions improves accuracy. In practice, recalibration should be estimated from related benchmark datasets wherever possible. Without such benchmark data, LLM predictions should be taken as estimates of relative effect size. Our recalibration method is tailored to our archive and may not generalize to substantially different experimental effects. More broadly, prompting and analytical choices (for example, simulating individual demographic profiles) may influence accuracy in non-obvious ways (Supplementary Information section 3.2), warranting systematic evaluation⁶⁴. Third, we have focused on estimating treatment effects on the studies' original outcome scales, rather than on standardized scales (for example, Cohen's *d*), which are important for applications such as power analysis. Previous work shows that LLMs underestimate the variance of human responses⁵⁶, complicating standardized effect-size estimation and statistical inference. Fourth, black-box, proprietary models may limit reproducibility owing to opaque training data^{5,28,57}; future studies should assess reproducibility over time and across models, which ideally requires documented information about models' training-data cutoffs and continuous sources of experimental data that are outside of these models' training data. Without comprehensive validation, deploying these models risks scientific harm, including perpetuating bias⁶⁵, prioritizing ideas on the basis of unreliable predictions and incentivizing more predictable studies²⁸. This highlights the need for research on the long-term effects of AI use in science.

Ultimately, the ability to simulate experiments and generate relatively accurate predictions at low cost could advance scientific and practical goals. Using AI to augment and inform, rather than replace, human judgement^{22,63} can help the social and behavioural sciences benefit from emerging technologies while remaining grounded in the fields' collective values. Although we view this work as an initial investigation, we anticipate that future research, such as efforts to improve accuracy (for example, model finetuning), validate performance across contexts (non-US datasets, field experiments) and mitigate risks (reducing bias, modelling variance), will be critical for advancing the use of LLMs to meaningfully support theoretically grounded, reliable and practically useful social and behavioural science.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41586-026-10742-x>.

1. Bail, C. A. Can generative AI improve social science? *Proc. Natl Acad. Sci. USA* **121**, e2314021121 (2024).
2. Grossmann, I. et al. AI and the transformation of social science research. *Science* **380**, 1108–1109 (2023).
3. Luo, X. et al. Large language models surpass human experts in predicting neuroscience results. *Nat. Hum. Behav.* **9**, 305–315 (2025).
4. Crockett, M. & Messeri, L. Should large language models replace human participants? Preprint at *PsyArXiv* <https://doi.org/10.31234/osf.io/4zdx9> (2024).
5. Abdurahman, S. et al. Perils and opportunities in using large language models in psychological research. *PNAS Nexus* **3**, pgae245 (2024).
6. Argyle, L. P. et al. Out of one, many: using language models to simulate human samples. *Polit. Anal.* **31**, 337–351 (2023).
7. Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B. & Larson, J. M. Synthetic replacements for human survey data? The perils of large language models. *Polit. Anal.* **32**, 401–416 (2024).

8. Atari, M., Xue, M. J., Park, P. S., Blasi, D. & Henrich, J. Which humans? Preprint at *PsyArXiv* <https://doi.org/10.31234/osf.io/5b26t> (2023).
9. Dillion, D., Tandon, N., Gu, Y. & Gray, K. Can AI language models replace human participants? *Trends Cogn. Sci.* **27**, 597–600 (2023).
10. Filippas, A., Horton, J. J. & Manning, B. S. Large language models as simulated economic agents: what can we learn from homo silicus? In *Proc. 25th ACM Conference on Economics and Computation* 614–615 (ACM, 2024).
11. Binz, M. & Schulz, E. Using cognitive psychology to understand GPT-3. *Proc. Natl Acad. Sci. USA* **120**, e2218523120 (2023).
12. Bai, H., Voelkel, J. G., Muldowney, S., Eichstaedt, J. C. & Willer, R. LLM-generated messages can persuade humans on policy issues. *Nat. Commun.* **16**, 6037 (2025).
13. Hackenburg, K. et al. The levers of political persuasion with conversational artificial intelligence. *Science* **390**, eaea3884 (2025).
14. Meta Fundamental AI Research Diplomacy Team (FAIR) et al. Human-level play in the game of diplomacy by combining language models with strategic reasoning. *Science* **378**, 1067–1074 (2022).
15. Park, J. S. et al. Generative agents: interactive simulacra of human behavior. In *Proc. 36th Annual ACM Symposium on User Interface Software and Technology* (eds Follmer, S. et al.) 1–22 (ACM, 2023).
16. Schramowski, P., Turan, C., Andersen, N., Rothkopf, C. A. & Kersting, K. Large pre-trained language models contain human-like biases of what is right and wrong to do. *Nat. Mach. Intell.* **4**, 258–268 (2022).
17. Yin, Y., Jia, N. & Waksak, C. J. AI can help people feel heard, but an AI label diminishes this impact. *Proc. Natl Acad. Sci. USA* **121**, e2319112121 (2024).
18. Lehr, S. A., Caliskan, A., Liyanage, S. & Banaji, M. R. ChatGPT as research scientist: probing GPT's capabilities as a research librarian, research ethicist, data generator, and data predictor. *Proc. Natl Acad. Sci. USA* **121**, e2404328121 (2024).
19. Time-sharing Experiments for the Social Sciences. *TESS* www.tessexperiments.org (2026).
20. Coppock, A., Leeper, T. J. & Mullinix, K. J. Generalizability of heterogeneous treatment effect estimates across samples. *Proc. Natl Acad. Sci. USA* **115**, 12441–12446 (2018).
21. Chang, S., Kennedy, A., Leonard, A. & List, J. A. *12 Best Practices for Leveraging Generative AI in Experimental Research* (National Bureau of Economic Research, 2024).
22. Broska, D., Howes, M. & van Loon, A. The mixed subjects design: treating large language models as potentially informative observations. *Sociol. Methods Res.* **54**, 1074–1109 (2025).
23. Chu, J. Y. et al. Academics are more specific, and practitioners more sensitive, in forecasting interventions to strengthen democratic attitudes. *Proc. Natl Acad. Sci. USA* **121**, e2307008121 (2024).
24. DellaVigna, S. & Linos, E. RCTs to scale: comprehensive evidence from two nudge units. *Econometrica* **90**, 81–116 (2022).
25. Dreber, A. et al. Using prediction markets to estimate the reproducibility of scientific research. *Proc. Natl Acad. Sci. USA* **112**, 15343–15347 (2015).
26. Milkman, K. L. et al. Megastudies improve the impact of applied behavioural science. *Nature* **600**, 478–483 (2021).
27. Harding, J., D'Alessandro, W., Laskowski, N. G. & Long, R. AI language models cannot replace human research participants. *AI Soc.* **39**, 2603–2605 (2024).
28. Messeri, L. & Crockett, M. J. Artificial intelligence and illusions of understanding in scientific research. *Nature* **627**, 49–58 (2024).
29. Binz, M. et al. A foundation model to predict and capture human cognition. *Nature* **644**, 1002–1009 (2025).
30. Petrov, N. B., Serapio-García, G. & Rentfrow, J. Limited ability of LLMs to simulate human psychological behaviours: a psychometric analysis. Preprint at <https://arxiv.org/abs/2405.07248> (2024).
31. Tjautja, L., Chen, V., Wu, T., Talwalkar, A. & Neubig, G. Do LLMs exhibit human-like response biases? A case study in survey design. *Trans. Assoc. Comput. Linguist.* **12**, 1011–1026 (2024).
32. Chen, Y., Hu, Y. & Lu, Y. Predicting field experiments with large language models. Preprint at <https://arxiv.org/abs/2504.01167> (2025).
33. Wang, X. et al. Self-consistency improves chain of thought reasoning in language models. In *Proc. 11th International Conference on Learning Representations* (poster) (ICLR, 2023).
34. Arora, S. et al. Ask me anything: a simple strategy for prompting language models. In *Proc. 11th International Conference on Learning Representations* (ICLR, 2023).
35. Voronov, A., Wolf, L. & Ryabinin, M. Mind your format: towards consistent evaluation of in-context learning improvements. In *Findings of the Association for Computational Linguistics: ACL 2024* (eds Ku, L.-W. et al.) 6287–6310 (ACL, 2024).
36. Hou, B., O'Connor, J., Andreas, J., Chang, S. & Zhang, Y. Promptboosting: black-box text classification with ten forward passes. In *Proc. International Conference on Machine Learning* (eds Krause, A. et al.) 13309–13324 (PMLR, 2023).
37. Open Science Collaboration. Estimating the reproducibility of psychological science. *Science* **349**, aac4716 (2015).
38. Milkman, K. L. et al. Megastudy shows that reminders boost vaccination but adding free rides does not. *Nature* **631**, 179–188 (2024).
39. Milkman, K. L. et al. A 680,000-person megastudy of nudges to encourage vaccination in pharmacies. *Proc. Natl Acad. Sci. USA* **119**, e2115126119 (2022).
40. Zickfeld, J. H. et al. Effectiveness of ex ante honesty oaths in reducing dishonesty depends on content. *Nat. Hum. Behav.* **9**, 169–187 (2025).
41. Brookman, D. E., Kalla, J. L., Caballero, C. & Easton, M. Political practitioners poorly predict which messages persuade the public. *Proc. Natl Acad. Sci. USA* **121**, e2400076121 (2024).
42. DellaVigna, S. & Vivalt, E. Forecasting social science: evidence from 100 projects. *National Bureau of Economic Research* <https://doi.org/10.3386/w34493> (2025).
43. Allen, J., Watts, D. J. & Rand, D. G. Quantifying the impact of misinformation and vaccine-skeptical content on Facebook. *Science* **384**, eadk3451 (2024).
44. DellaVigna, S. & Pope, D. What motivates effort? Evidence and expert forecasts. *Rev. Econ. Stud.* **85**, 1029–1069 (2018).
45. Vlasceanu, M. et al. Addressing climate change with behavioral science: a global intervention tournament in 63 countries. *Sci. Adv.* **10**, ead5778 (2024).
46. Tappin, B. M., Wittenberg, C., Hewitt, L. B., Berinsky, A. J. & Rand, D. G. Quantifying the potential persuasive returns to political microtargeting. *Proc. Natl Acad. Sci. USA* **120**, e2216261120 (2023).
47. Voelkel, J. G. et al. Megastudy testing 25 treatments to reduce antidemocratic attitudes and partisan animosity. *Science* **386**, eadh4764 (2024).
48. Voelkel, J. G. et al. A registered report megastudy on the persuasiveness of the most-cited climate messages. *Nat. Clim. Chang.* **16**, 214–225 (2026).
49. Saccardo, S. et al. Field testing the transferability of behavioural science knowledge on promoting vaccinations. *Nat. Hum. Behav.* **8**, 878–890 (2024).
50. Mason, K. et al. A megastudy of behavioral interventions to increase voter registration ahead of the 2024 US presidential election. Preprint at *PsyArXiv* https://doi.org/10.31234/osf.io/p6b2e_v3 (2025).
51. Goldwert, D. et al. A megastudy of behavioral interventions to catalyze public, political, and financial climate advocacy. *PNAS Nexus* **5**, pgaf400 (2026).
52. Druckman, J. N. *Experimental Thinking* (Cambridge Univ. Press, 2022).
53. Freese, J. & Peterson, D. Replication in social science. *Annu. Rev. Sociol.* **43**, 147–165 (2017).
54. Bai, X., Wang, A., Sucholutsky, I. & Griffiths, T. L. Explicitly unbiased large language models still form biased associations. *Proc. Natl Acad. Sci. USA* **122**, e2416228122 (2025).
55. Wang, A., Morgenstern, J. & Dickerson, J. P. Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nat. Mach. Intell.* **7**, 400–411 (2025).
56. Park, P. S., Schoenegger, P. & Zhu, C. Diminished diversity-of-thought in a standard large language model. *Behav. Res. Methods* **56**, 5754–5770 (2024).
57. Frank, M. C. Openly accessible LLMs can help us to understand human cognition. *Nat. Hum. Behav.* **7**, 1825–1827 (2023).
58. Wack, M., Ehrett, C., Linvill, D. & Warren, P. Generative propaganda: evidence of AI's impact from a state-backed disinformation campaign. *PNAS Nexus* **4**, pgaf083 (2025).
59. Response retrieved from ChatGPT-4 on 22 May. *ChatGPT* <https://chatgpt.com/share/2a2b66d0-e9fd-421f-bedd-bb6f9354742a> (2024).
60. Almaatouq, A. et al. Beyond playing 20 questions with nature: integrative experiment design in the social and behavioral sciences. *Behav. Brain Sci.* **47**, e33 (2024).
61. Zhu, J.-Q., Xie, H., Arumugam, D., Wilson, R. C. & Griffiths, T. L. Using reinforcement learning to train large language models to explain human decisions. Preprint at <https://arxiv.org/abs/2505.11614> (2025).
62. Tyner, A. H. et al. Investigating the replicability of the social and behavioural sciences. *Nature* **652**, 143–150 (2026).
63. Cremer, D. D. & Kasparov, G. AI should augment human intelligence, not replace it. *Harvard Business Review* <https://hbr.org/2021/03/ai-should-augment-human-intelligence-not-replace-it> (2021).
64. Cummins, J. The threat of analytic flexibility in using large language models to simulate human data: a call to attention. Preprint at <https://arxiv.org/abs/2509.13397> (2025).
65. Obermeyer, Z., Powers, B., Vogeli, C. & Mullainathan, S. Dissecting racial bias in an algorithm used to manage the health of populations. *Science* **366**, 447–453 (2019).

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

© The Author(s), under exclusive licence to Springer Nature Limited 2026

Methods

Primary archive

Our primary testing archive of survey experimental results is composed of 50 experiments conducted between 2016 and 2022 by TESS, an NSF-sponsored programme in which researchers—typically social scientists—propose survey-based experiments that, if selected through a peer-review process, will be run on nationally representative probability samples. We supplemented the archive with 20 experiments reported in a meta-analysis of two recent replication projects^{66,67}. The meta-analysis tested experimental results obtained from probability samples (studies conducted before 2012 largely by means of TESS) in new experiments they conducted on convenience samples (that is, Amazon MTurk) and found similar results across the two types of samples. We included data collected through both sample sources to maximize statistical power. TESS requires study authors to submit proposals stating their hypotheses and justifying their sample size, which are then peer-reviewed before the experiment is fielded on nationally representative samples, effectively providing us with an archive of preregistered, highly powered, high-quality experiments with openly posted stimuli and data. We obtained the final archive of 70 experiments after excluding studies that did not meet our inclusion criteria of experiments (1) that target the general US population, (2) that do not use a conjoint or list design, (3) whose stimuli were primarily or entirely text based and (4) wherein randomization did not depend on participants' responses to questions. We considered studies to be text based if they primarily relied on text and any information communicated by means of images was redundant with the text. For each study in the archive, we included up to 12 conditions and three outcome items corresponding to the first three outcome variables measured post manipulation. In addition to coding the original studies' manipulation text and outcome questions, the authors along with a team of eight research assistants additionally coded metadata, including the study's hypotheses, discipline(s), topic and publication status, by means of a process involving several rounds of verification to ensure high accuracy. We considered unpublished studies and studies published or posted as preprints after September 2021 unlikely to have been included in GPT-4's training data, based on archived OpenAI documentation (<https://archive.is/WMcad#selection-1835.0-1835.5>) (archived 23 November 2024, after we recorded LLM responses) and GPT-4's technical report⁶⁸. See 'Primary experimental archive details' in the Supplementary Information for more information on this archive, including our criteria for including experiments, experimental conditions and outcome items and the coding process; see Supplementary Table 3 for a full list of its experiments.

This primary testing archive has several strengths. First, the experiments are high quality: all are highly statistically powered, peer reviewed and conducted on nationally representative samples, with preregistered study designs and materials, as well as openly accessible materials and data. The use of nationally representative samples of American individuals is particularly valuable, allowing us to assess the accuracy of LLM-derived predictions for demographic subgroups. Second, the archive is substantively broad and diverse. The experiments were designed by 77 social and behavioural scientists from a range of fields (including political science, psychology, sociology, social policy, public health, communication) and test the effects of many different types of experimental treatments (such as framing effects, salience of topics, priming social identities) on a range of outcomes (such as political, cultural and religious attitudes, prejudice towards minorities, happiness) (see Supplementary Table 3 for the full list of experiments). Third, results for many experiments were not published or posted publicly by the end of GPT-4's training-data window, allowing us to specifically test for LLMs' predictive capacity on experiments that the model could not have been exposed to. Fourth, rather than rely on the original researchers' analyses, we implemented a consistent

analytic approach to estimate experimental treatment effects. Doing so allowed us to avoid researcher bias and also estimate all possible experimental contrasts, including those effects not hypothesized by the original researchers and therefore unlikely to have been reported in published or publicly posted papers. Finally, we collected forecasts from 2,659 workers on Prolific (49.4% women, 48.9% men, 1.6% other; $M_{\text{age}} = 41.73$; 70.4% white) for all the experiments in our primary test archive (see 'Layperson Forecasts' in the Supplementary Information and Supplementary Fig. 1), which provided us with a benchmark against which we compared LLM performance.

The primary testing archive also has important limitations. Most critically, it only features studies representative of the US population. Also, although it includes studies from a diversity of disciplines, many disciplines are not included (for example, cognitive psychology, behavioural economics, development economics, marketing). Finally, the archive consists entirely of survey experiments with text-based stimuli and self-reported dependent measures and does not include field experiments, behavioural dependent variables or image or video stimuli.

Primary archive: analytical approach

In the primary archive, we used a consistent analytical approach across experiments. To assess predictive accuracy, we (1) randomly selected one condition per study to serve as the reference condition against which all others were compared (and one dependent variable, in cases where the original study contained several); (2) computed the LLM-predicted treatment effects using the approach described above; (3) correlated the resulting predicted treatment effects with actual effects estimated using the original data; and (4) repeated this process 32 times (to account for randomness in step 1), recording the median correlation coefficient. Note that we were unable to preregister our analytic approach because of our familiarity with the analysed test archives, but we present several robustness check analyses (Supplementary Information sections 2.1, 2.2 and 3.1).

In addition to raw correlations (r), we report adjusted correlations (r_{adj}) that adjust for statistical uncertainty in estimates of original treatment effects. To compute these error-adjusted correlations, we used a two-stage random effects meta-analysis to account for the statistical uncertainty in the treatment effects from each original study. For the first stage, we estimated treatment effects using ordinary least squares regression, extracting the corresponding covariance matrix of the sampling errors. For the second stage, we assumed that predicted and true effects follow a bivariate normal distribution and estimated the parameters of this distribution incorporating the aforementioned sampling errors using the metafor R package⁶⁹. The estimated correlation parameter ρ of this distribution, and its 95% CI, is the disattenuated or adjusted correlation (r_{adj}). We also calculated the absolute accuracy of predictions by estimating the r.m.s.e., adjusting for the uncertainty in treatment effects (θ) using the following formula, where i indexes each treatment effect, x_i and y_i represent the predicted and observed effect size, respectively:

$$\text{Adjusted r.m.s.e.} = \sqrt{\frac{1}{n} \sum_i (y_i - x_i)^2 - \theta_i^2} \quad (1)$$

Secondary archive of intervention megastudies

We additionally analysed 15 recent, large-scale, many-treatment experiments testing interventions crafted by researchers from an array of disciplines including psychology, political science, marketing, and economics, sociology, organizational behaviour, as well as non-academic practitioners. We included megastudies targeting socially beneficial outcomes that were conducted on sufficiently large samples: (1) Allen et al.⁴³ (2024; $n = 9,228$), testing the effects of 90 messages on vaccination intentions; (2) Voelkel et al.⁴⁸ (2026; $n = 10,638$), testing the

Article

effects of 10 widely cited treatments on climate change attitudes; (3) DellaVigna and Pope⁴⁴ (2018, $n = 9,321$), testing 15 treatments targeting effortful task behaviour; (4) Vlasceanu et al.⁴⁵ (2024, $n = 6,735$), testing 11 text treatments on climate change attitudes; (5) Tappin et al.⁴⁶ (2023, $n = 62,738$), testing 59 treatments targeting universal basic income and immigration policy attitudes; (6) Zickfeld et al.⁴⁰ (2025, $n = 20,553$), testing 21 treatments targeting tax compliance; (7) Mason et al.⁵⁰ (2025, $n = 11,733$), testing 10 treatments targeting voting intent; (8) Goldwert et al.⁵¹ (2026, $n = 27,343$), testing 17 treatments targeting climate action; (9) Voelkel et al.⁴⁷ (2024, $n = 25,121$), testing 25 treatments targeting political beliefs; (10) Broockman et al.⁴¹ (2024, $n = 61,869$), testing 172 treatments targeting various political attitudes; (11) Saccardo et al.⁴⁹ (2024, $n = 259,249$), testing 14 treatments targeting COVID-19 vaccine booster uptake; (12) Milkman et al.³⁹ (2022, $n = 662,170$), testing 22 text message treatments on behavioural vaccine uptake; (13) DellaVigna and Linos²⁴ (2022, $n = 960,472$), a meta-analysis of 14 nudge treatments targeting several behavioural outcomes; (14) Milkman et al.³⁸ (2024, $n = 2,677,402$), testing seven treatments targeting COVID-19 vaccine booster uptake; and (15) Milkman et al.²⁶ (2021, $n = 57,790$), testing 53 28-day-long treatments targeting behavioural gym attendance. See ‘Secondary archive of megastudies, Expert forecasts’ in the Supplementary Information and Supplementary Table 4 for more information.

Secondary archive: analytical approach

Given that the megastudies always included a clearly defined control group against which the effects of all treatments were computed, unlike in the primary archive, our goal here was to assess how accurately our LLM-based approach can distinguish the most and least effective interventions in a megastudy. For this reason, we estimated the correlation for all treatment effects (relative to the control condition) in a single study and then calculated a meta-analytic mean of these correlations. We computed adjusted correlations (r_{adj}) using the approach described above for the primary archive. Note that r.m.s.e. is not reported for this archive because the wide variation in outcome variable scales across experiments (for example, Likert scale measure of climate change attitudes versus number of days someone went to the gym) makes it an unreliable metric for cross-experimental comparison.

LLM prompting

Our approach involved directly prompting the LLM application programming interface, rather than using commercial web chat applications (such as ChatGPT). ChatGPT—a chatbot designed for broad commercial use—has further capabilities, such as internet access, that the GPT-4 model itself does not have. LLM prompts were administered in unique sessions with no memory across sessions, ensuring each response was generated independent of the others. Automating the process of obtaining and recording LLM responses also helped us avoid manual coding errors. Our primary analysis used GPT-4, which we estimated to be the most advanced model when we first submitted this paper. Owing to practical constraints such as rate limits and costs, we restricted our analysis to this one proprietary model. In addition to GPT-4, for our main analyses, we also used prominent open-weight models Gemma-3, GPT-OSS and DeepSeek v.3, and we believe that these models’ responses will remain reproducible even if proprietary models are changed in non-transparent ways.

Broadly speaking, to achieve the goal of predicting experimental results, LLMs could be prompted to either (1) directly predict experimental results, similar to what forecasters are required to do; or (2) simulate individual participants’ responses to experimental stimuli, similar to what research participants do. Here we adopted the latter strategy because simulation more closely mirrors the original data-generating process and avoids the meta-level judgements required by direct forecasting. This approach also allowed us to examine accuracy for demographic subgroups. Note that for each prompt, we sampled five LLM responses and averaged them.

In all prompts, we ordered the scale labels of outcome scales such that higher values on the scale were paired with labels indicating positive valence or higher degree of the measured quantity (for example, 1 = not at all to 5 = very much, rather than 1 = very much to 5 = not at all). Although most original treatments in the secondary archive relied on text, some treatments had to be converted to static text (such as transcripts of videos and high-level descriptions of image content) before being added to prompts. For megastudies with behavioural outcomes (for example, vaccine uptake), the prompt asked how likely the hypothetical participant would be to engage in the particular behaviour. See ‘Prompting strategy’ in the Supplementary Information for further details.

Survey of social scientists

To obtain a large and diverse sample of social scientists, in May 2025, we distributed the survey by means of email to researchers who (1) have submitted experimental ideas to TESS, (2) were contacted for previous forecasting or metascience projects (for example, ref. 70) and (3) are members of the Society for Judgment and Decision Making listserv. Our sample included 460 participants (40.84% women, 20.95% non-white) representing several social science disciplines (including 33.5% psychologists, 31.3% political scientists and 17.6% sociologists, among others) and career stages (including 20.2% professors, 20.2% associate professors, 21.3% assistant professors, 16.3% graduate students and 11.5% postdocs, among others). Further details about the sample, recruitment strategy and a list of all questions can be found in the Supplementary Information.

Ethics statement

This research was approved by Stanford University’s Institutional Review Board. All procedures were conducted in accordance with applicable guidelines and regulations, including obtaining informed consent from participants in both the forecasting survey and the survey of social scientists.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data used in this paper are available at <https://codeocean.com/capsule/9843791/tree/v1>. All study materials are available in the Supplementary Information. Source data are provided with this paper.

Code availability

Code to reproduce the results is available at <https://codeocean.com/capsule/9843791/tree/v1>. We also invite researchers to try the model in our web demo (<https://treatmenteffect.app/>), which generates AI-based forecasts of experimental effects using the approach detailed in this paper.

66. Coppock, A. Generalizing from survey experiments conducted on mechanical turk: a replication approach. *Political Sci. Res. Methods* **7**, 613–628 (2019).
67. Mullinix, K. J., Leeper, T. J., Druckman, J. N. & Freese, J. The generalizability of survey experiments. *J. Exp. Political Sci.* **2**, 109–138 (2015).
68. Achiam, J. et al. GPT-4 technical report. Preprint at <https://arxiv.org/abs/2303.08774> (2023).
69. Viechtbauer, W. metafor: Meta-Analysis Package for R v.4.8-0 (2015).
70. Rios, K., Roth, Z. C. & Coleman, T. J. III. The importance of scientists’ intellectual humility for communicating effectively across ideological and identity-based divides. *Proc. Natl Acad. Sci. USA* **122**, e2400930121 (2025).

Acknowledgements We thank N. Egan, K. Fuller, V. Isarraras, S. Kane, B. Kinder, A. Kissi, K. McPeck, S. Muldowney, C. Redekopp and T. Ren for research assistance. We are grateful to researchers who shared data, including J. Chu, S. DellaVigna, J. Druckman, A. Duckworth, D. Goldwert, N. Malhotra, K. Mason, K. Milkman, K. Okst, S. Saccardo, M. Vlasceanu and J. Voelkel. We also thank J. Allen, M. Atari, C. Bail, M. Bernstein, M. Dehghani, D. Broockman,

K. Clayton, M. Crockett, D. Ganguli, D. Dillon, J. Eichstaedt, M. Frank, N. Goodman, D. Green, J. Kalla, P. Liang, J. McClelland, D. Rand, R. Orchinik, N. Otis, S. Rathje, B. Tappin, A. Van Loon, A. Waytz and D. Willner for invaluable feedback on the project.

Author contributions L.H., A.A. and R.W. designed the research. A.A., L.H., I.G. and R.W. assembled the experimental archives. L.H. and A.A. analysed the data. A.A., L.H. and R.W. wrote the paper. L.H. built the web tool.

Funding We acknowledge funding from the Stanford Center on Philanthropy and Civil Society.

Competing interests The authors declare no competing interests.

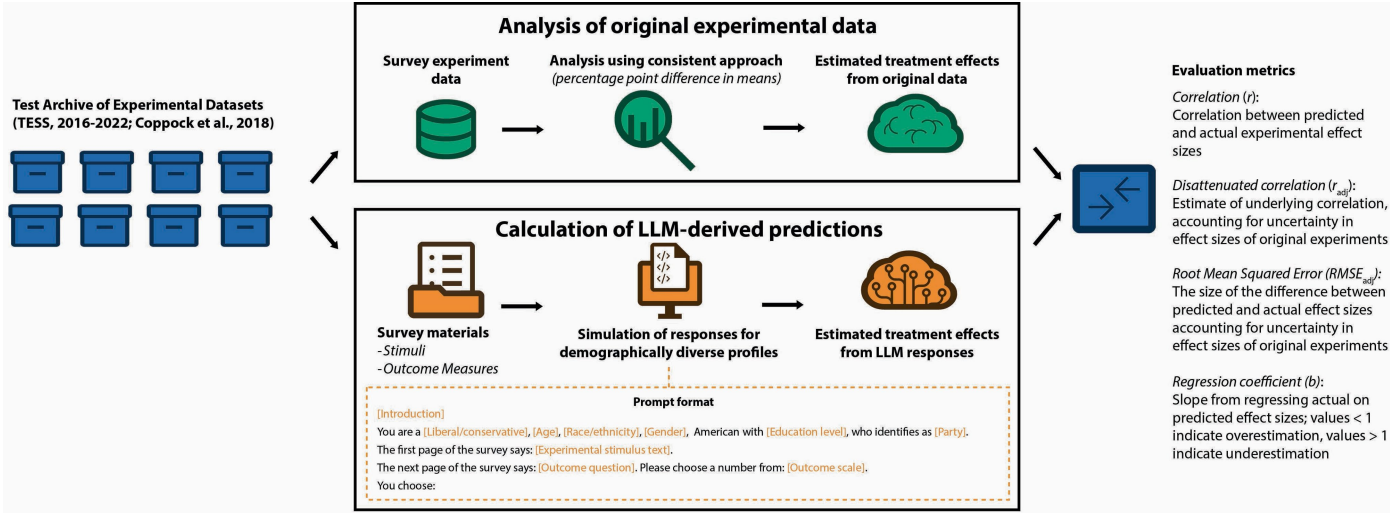
Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41586-026-10742-x>.

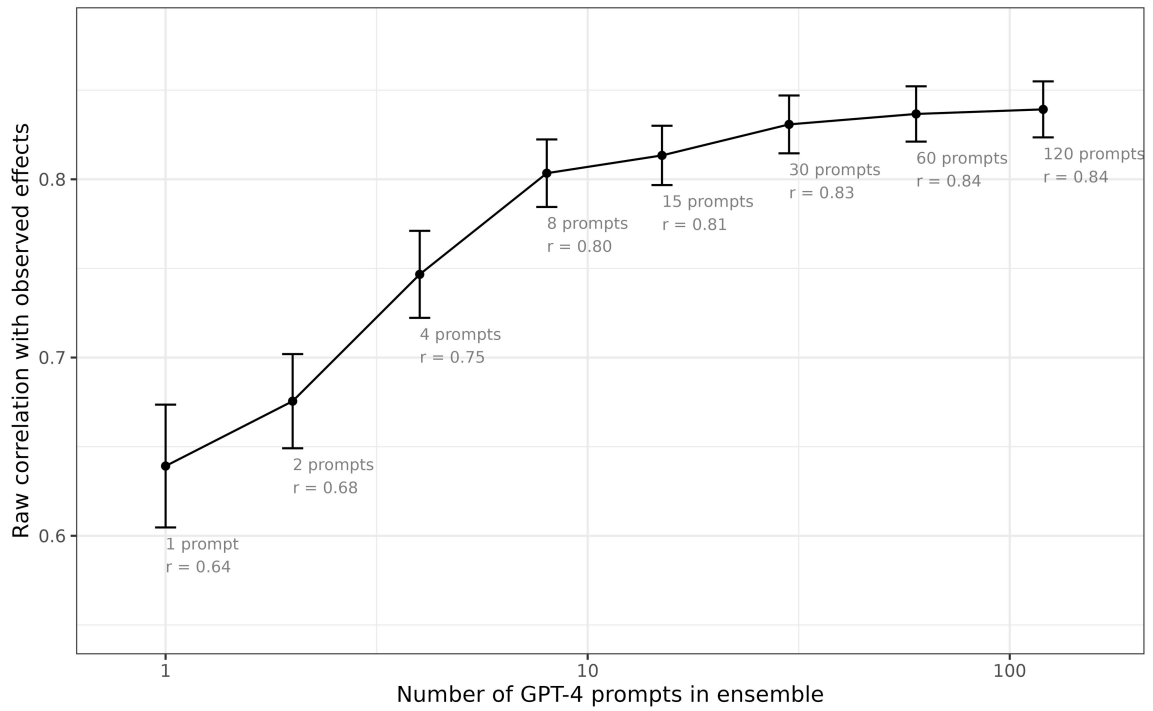
Correspondence and requests for materials should be addressed to Ashwini Ashokkumar, Luke Hewitt or Robb Willer.

Peer review information *Nature* thanks Jamie Cummins, James Druckman, John Protzko and Youyou Wu for their contribution to the peer review of this work. Peer reviewer reports are available.

Reprints and permissions information is available at <http://www.nature.com/reprints>.

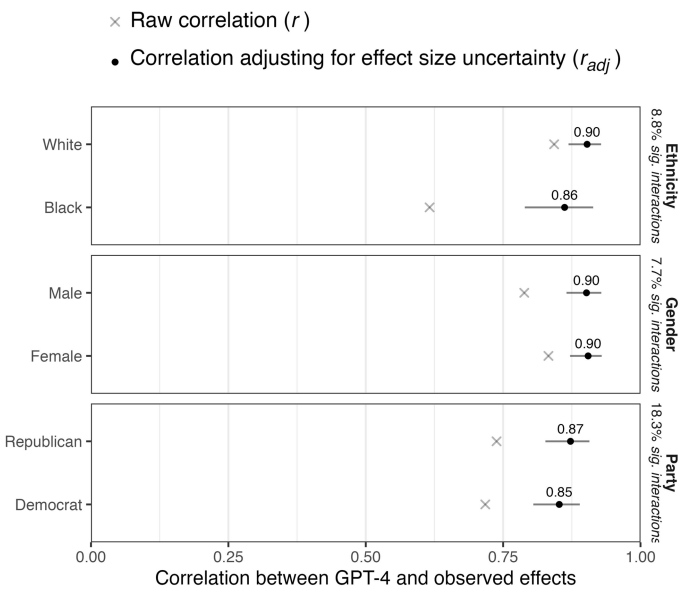


Extended Data Fig. 1 | Method. We re-analyze raw data from 70 nationally-representative U.S. studies and estimate treatment effects in a consistent manner. We use a Large Language Model to predict results of those same experiments, providing the original study materials and demographically-diverse participant profiles, and then calculate the average predicted response for each condition. We evaluate the accuracy of the model in terms of the correspondence between observed- and predicted- treatment effects.



Extended Data Fig. 2 | Averaging across more LLM prompts improves predictive accuracy. Error bars indicate 95% confidence intervals for the mean Pearson correlation across 32 simulation runs. Each run uses a different

random subset of prompts (drawn without replacement from 120). The CI reflects uncertainty in the mean across simulation runs, but does not include uncertainty over the sample of 70 experiments or the sample of 120 prompts.



Extended Data Fig. 3 | The accuracy of LLM-derived predictions for demographic subgroups in the US. Error bars indicate 95% confidence intervals.

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☐	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☐	☒ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☐	☒ A description of all covariates tested
☐	☒ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☐	☒ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☐	☒ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☐	☒ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	Our main analyses involve secondary archives of experiments run by a large number of social scientists. We collected two primary datasets. First, we collected layperson forecasts via Prolific. Second, we collected survey data from 460 social scientists, for which we used Qualtrics.
Data analysis	Data were analyzed using R. All relevant code is available in our repository.

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Data, code, and materials used in the analysis are available in the SI and in our repository.. The manuscript includes a Data Availability Statement.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender	We examined whether LLM-derived predictions of experimental results vary in accuracy for different demographic subgroups within the US. Here, we examined whether predictive accuracy was different for different genders. We report gender information for our survey of social scientists.
Reporting on race, ethnicity, or other socially relevant groupings	We examined whether LLM-derived predictions of experimental results vary in accuracy for different demographic subgroups within the US. Here, we examined whether predictive accuracy was different for different ethnic groups. We also report ethnicity information for our survey of social scientists.
Population characteristics	The test archives we used contain secondary data.
Recruitment	The test archives we used contain secondary data. For the primary data we collected (layperson forecasts and survey of social scientists), we report information on recruitment in the manuscript and SI.
Ethics oversight	Stanford IRB

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☐ Life sciences ☒ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see nature.com/documents/nr-reporting-summary-flat.pdf

Behavioural & social sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Analysis of secondary archives of social science experiments
Research sample	Our main datasets are archives of social science experiments. We additionally collected layperson forecasts from Prolific, and we surveyed 460 social scientists (convenience sample).
Sampling strategy	For our sample of social scientists, we used convenience sampling. For collecting layperson forecasts, we used Prolific.
Data collection	We used existing archives of experiments.
Timing	NA
Data exclusions	NA
Non-participation	NA
Randomization	NA

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.